

Einstein's General Theory of Relativity

Lecture 11: Rindler Horizons and the Schwarzschild Metric

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Chapter 1

Lecture 11: Rindler Horizons and the Schwarzschild Metric

Overview. Topology, curvature, and coordinates are easy to confuse, so they must first be kept separate. Gauss–Bonnet fixes the total curvature of a closed surface from its number of handles, yet a suitable metric can make a torus flat, and deleting one point from a sphere allows a flat metric pulled back from the plane. Normal coordinates then remove the connection at one event without removing curvature. With that distinction in hand, proper time becomes the action for a free particle, its weak-field limit identifies g_{00} with the Newtonian potential, and uniformly accelerated observers provide a flat-spacetime model of a horizon. Rindler coordinates make the model exact. The closing construction uses spherical symmetry, the Newtonian limit, and Birkhoff’s theorem to introduce the Schwarzschild exterior metric.

We use the spacetime signature

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1) \quad (1.1)$$

and write $ds^2 = c^2 d\tau^2$ for a timelike worldline. Factors of c are retained when they clarify dimensions and suppressed where the geometry is easier to read in units with $c = 1$.

1.1 Topology Fixes Total Curvature

On the outer rim of an ordinary torus the Gaussian curvature K is positive; on the inner rim it is negative. The two contributions cancel when integrated over the whole surface. A deformation into a coffee-cup shape can move positive and negative curvature from place to place, but it cannot alter the total. For a closed orientable surface Σ_g of genus g , Gauss–Bonnet states

$$\boxed{\int_{\Sigma_g} K \, dA = 2\pi\chi(\Sigma_g) = 4\pi(1 - g).} \quad (1.2)$$

Thus, after division by 4π , the sphere has total $+1$, the torus has 0 , a two-handled surface has -1 , and a three-handled surface has -2 .

The theorem gives an obstruction. A sphere or a surface with two or more handles cannot carry an everywhere-flat metric, because $K = 0$ everywhere would force the integral to vanish. A torus passes this necessary test.

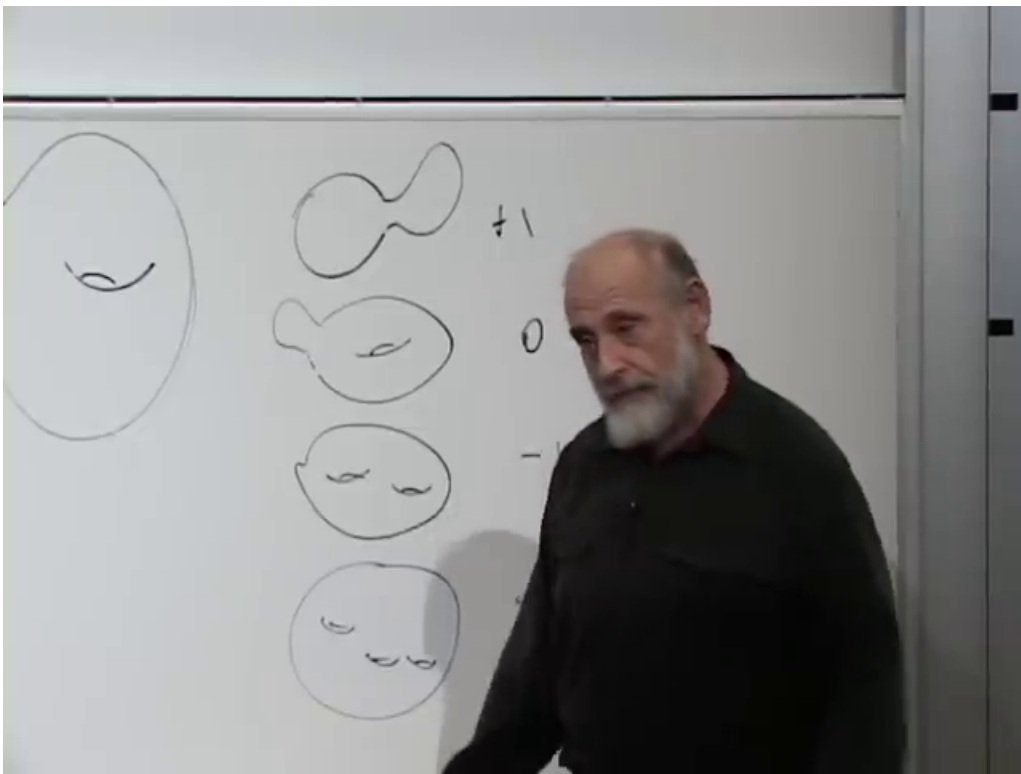


Figure 1.1: The board classification of closed surfaces by handles. In the normalization used in the discussion, their integrated curvatures are $+1, 0, -1, -2, \dots$

Lecture frame: 00:03:25.

Question from the audience. If the torus has zero integrated curvature, does that prove that the torus already drawn on the board can be unfolded into a flat sheet? ^a

Response. No. A zero integral says only that positive and negative curvature cancel. It does not force the particular metric to have $K = 0$ point by point. Total curvature is a topological invariant; local curvature still depends on the metric.

^aLecture timestamp: 00:01:47–00:02:16.

Question from the audience. Can one nevertheless put an everywhere-flat metric on a torus? ^a

Response. Yes. Begin with a Euclidean rectangle and identify opposite edges. The quotient is a torus and inherits the flat metric locally. It cannot be embedded in ordinary three-dimensional Euclidean space as the familiar doughnut without bending, but intrinsic flatness does not require that embedding.

^aLecture timestamp: 00:03:42–00:04:15.

This is the first distinction to keep: topology constrains which metrics are possible, but topology is not itself a metric.

1.2 A Sphere with One Point Removed

Delete the north pole N from a sphere. The remaining surface is topologically a plane. To see it, rest the sphere on a plane at its south pole and draw a line from N through each point P of the sphere. Continue the line until it meets the plane. This stereographic projection is one-to-one from $S^2 \setminus \{N\}$ to $\mathbb{R}^2$; the missing north pole corresponds to infinity.

Now place the ordinary Euclidean metric

$$d\ell^2 = du^2 + dv^2 \tag{1.3}$$

on the plane and pull it back through the projection. The resulting metric on the punctured sphere is flat. It is not the standard round metric. In coordinates that remain finite on the drawn sphere, its components diverge near N , because a short visual displacement there represents a very large Euclidean displacement in the plane. Measuring rods adapted to this metric shrink as they approach the missing point, and geodesic triangles have angle sum 180° .

Question from the audience. Does removing one point really let the sphere become flat, and what becomes singular? ^a

Response. The punctured surface can be assigned the metric pulled back from the plane, so its Gaussian curvature is zero everywhere that remains. The deleted point lies at infinite distance in that metric. What diverges near it are coordinate components of the pulled-back metric, not the intrinsic curvature on the punctured surface.

^aLecture timestamp: 00:04:17–00:07:55.

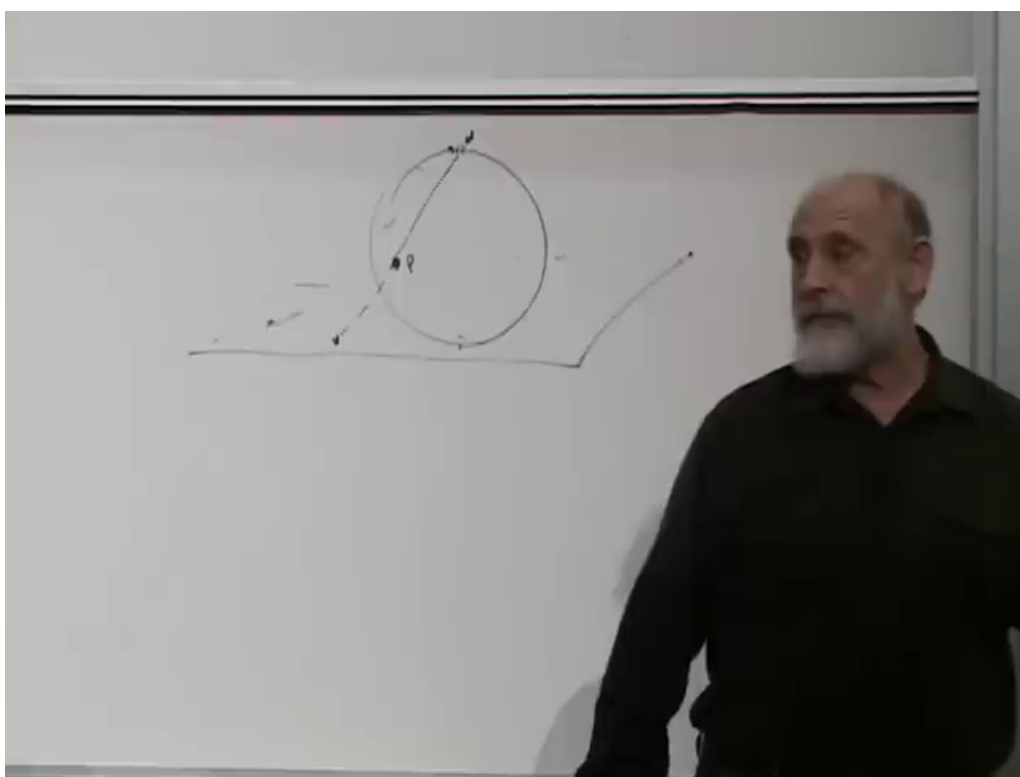


Figure 1.2: Stereographic projection from the deleted north pole onto a tangent plane. Points approaching the deleted pole are sent to arbitrarily large distance.

Lecture frame: 00:07:48.

Question from the audience. Are the geodesics of this flat metric still great circles of the round sphere? ^a

Response. In general, no. Geodesics are determined by the chosen metric. Here they are inverse images of straight lines in the plane. Under stereographic projection those inverse images are circles through the deleted north pole, with a limiting straight-line case, rather than arbitrary great circles of the round metric.

^aLecture timestamp: 00:07:55–00:08:25.

Question from the audience. Does this kind of two-variable mapping suggest a direct physical application? ^a

Response. No immediate example is identified. The question is left open rather than used to infer a physical claim, and the discussion moves to what mappings between different dimensions can mean mathematically.

^aLecture timestamp: 00:08:25–00:08:58.

1.2.1 Why cardinality is not a coordinate transformation

The projection prompts a broader question: can spaces of different dimension be put into one-to-one correspondence? As sets, the answer can be yes. For $x, y \in (0, 1)$, choose decimal expansions that do not end in an infinite string of nines,

$$x = 0.x_1x_2x_3 \dots, \quad y = 0.y_1y_2y_3 \dots,$$

and interleave their digits:

$$F(x, y) = 0.x_1y_1x_2y_2x_3y_3 \dots \quad (1.4)$$

The alternating digits can be separated again, so this construction encodes a pair of real numbers in one real number. The convention on terminating expansions removes the familiar ambiguity $0.5000 \dots = 0.4999 \dots$.

Question from the audience. If two coordinates can be encoded in one number, why can this not serve as a change of coordinates? ^a

Response. Because a coordinate transformation in geometry must at least be continuous, and for tensor calculus it must be differentiable with a nonsingular Jacobian. Digit interleaving has none of those properties. Infinitesimally close points can be sent far apart, and a smooth function of x and y becomes violently irregular as a function of F . Equality of cardinality does not preserve dimension, neighborhoods, or geometry.

^aLecture timestamp: 00:08:58–00:15:40.

1.3 Normal Coordinates Remove the Connection, Not Curvature

At any event P , geodesic or Riemann normal coordinates can be chosen so that

$$g_{\mu\nu}(P) = \eta_{\mu\nu}, \quad \partial_\alpha g_{\mu\nu}(P) = 0, \quad \Gamma_{\mu\nu}^\rho(P) = 0. \quad (1.5)$$

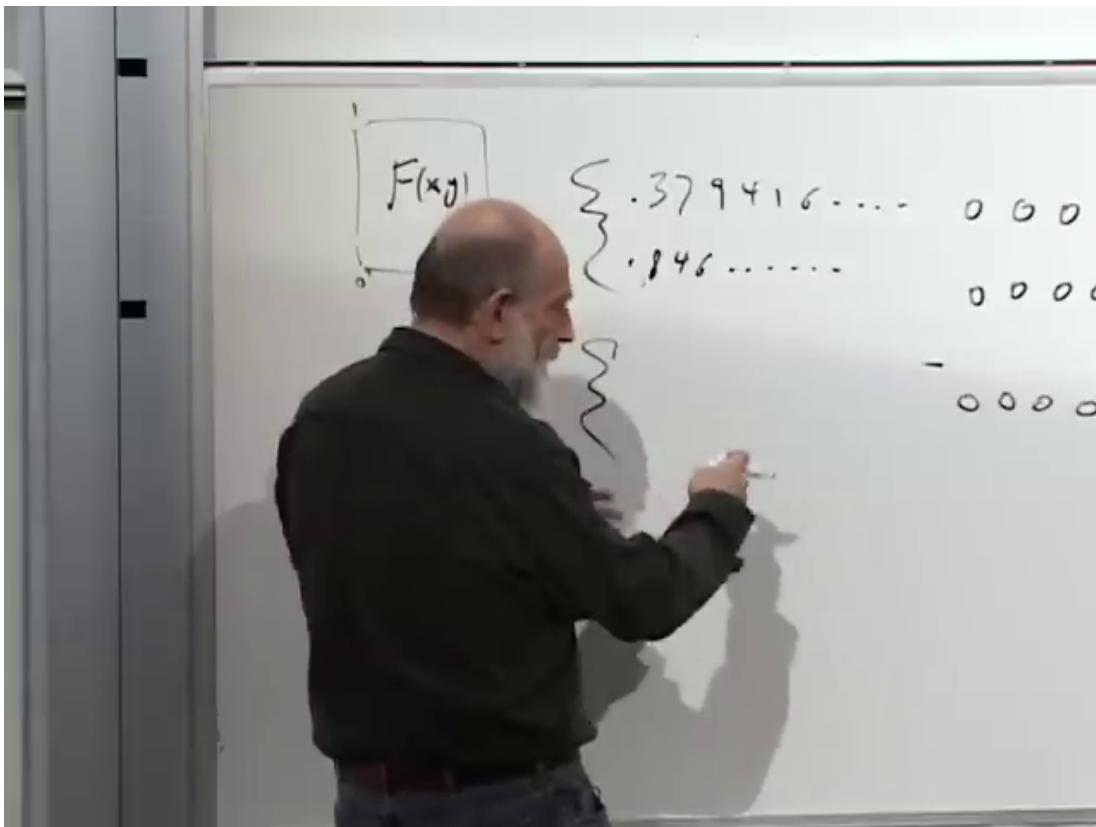


Figure 1.3: The digit-interleaving construction used to compare the cardinality of a square with that of an interval.

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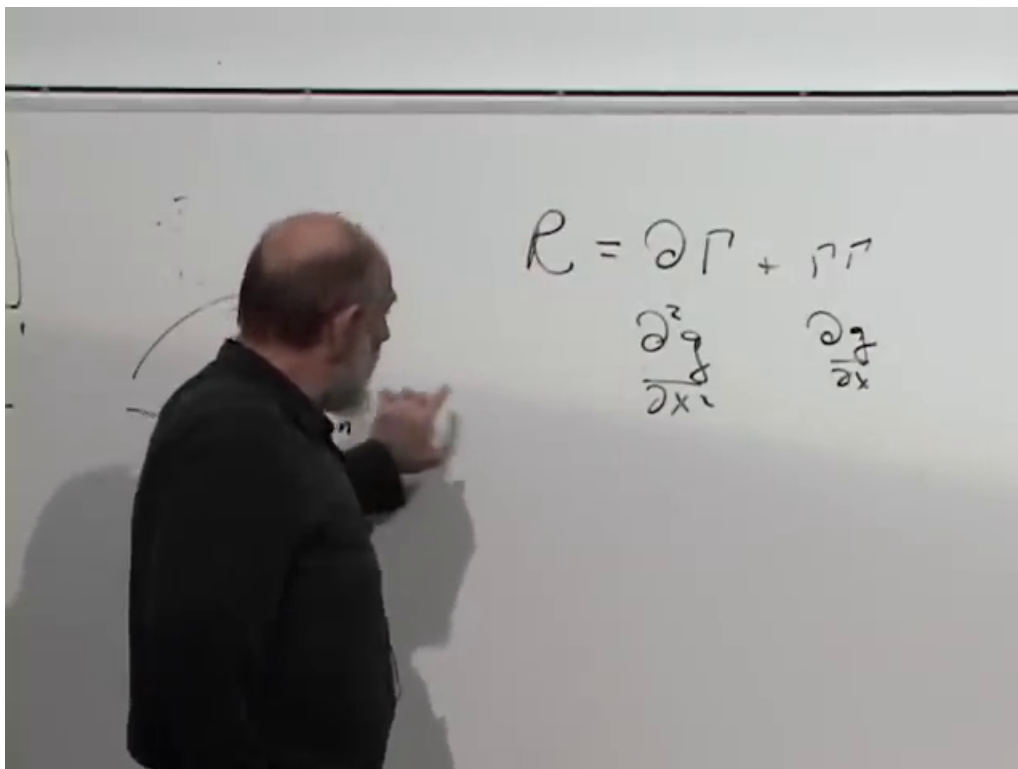


Figure 1.4: The board isolates curvature as $\partial\Gamma + \Gamma\Gamma$, and therefore as second derivatives of the metric after first derivatives have been removed at one event.

Lecture frame: 00:19:35.

This is the precise local content of freely falling coordinates. It does not imply that the derivatives of the connection vanish. Since

$$\Gamma \sim g^{-1}\partial g,$$

one has schematically

$$R \sim \partial\Gamma + \Gamma\Gamma \sim \partial^2 g + (\partial g)^2. \quad (1.6)$$

At P , the quadratic term vanishes in normal coordinates while the second-derivative term generally survives.

For example, with the convention

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \dots,$$

the lowered tensor at P is a linear combination of second metric derivatives,

$$R_{\alpha\beta\gamma\delta}(P) = \frac{1}{2} (g_{\alpha\delta,\beta\gamma}g_{\beta\gamma,\alpha\delta} - g_{\alpha\gamma,\beta\delta} - g_{\beta\delta,\alpha\gamma})_P. \quad (1.7)$$

Question from the audience. If the metric is Minkowskian and the connection vanishes at P , how can curvature remain there? ^a

Response. The coordinate choice fixes the value and first derivatives of the metric at one event, not its second derivatives. Curvature measures precisely the obstruction that remains at that order. One can remove a gravitational field locally in the sense of free fall, but one cannot transform away tidal effects in a genuinely curved spacetime.

^aLecture timestamp: 00:15:42–00:19:55.

1.4 Geodesics and the Proper-Time Action

A geodesic is defined by parallel transport of its tangent:

$$u^\nu \nabla_\nu u^\mu = 0, \quad u^\mu = \frac{dx^\mu}{d\tau}. \quad (1.8)$$

In affine coordinates this becomes

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (1.9)$$

The variational description is local. A timelike geodesic locally maximizes proper time among nearby timelike curves, up to conjugate points. A Riemannian geodesic is a stationary curve. A sufficiently short segment locally minimizes length, but a longer segment may lose that property after conjugate points and become a saddle; it need not be the global shortest route.

Question from the audience. Is a geodesic always the shortest path, or in spacetime always the longest elapsed time? ^a

Response. The invariant statement is stationarity under small variations. Timelike geodesics locally maximize proper time over a suitable interval. In a positive-definite metric a sufficiently short geodesic segment is a local length minimum. A longer geodesic can become unstable or saddle-like; global claims require additional conditions and can fail when several geodesics join the same points.

^aLecture timestamp: 00:20:03–00:22:08.

Imagine two points on opposite sides of a symmetric hill. A taut string can settle along either side, giving two locally stable geodesics. A perfectly centered path over the summit is also stationary. Within the smooth family pictured on the board it sits at the top of the route-length comparison, but under unrestricted path variations it is safer to call it unstable: the smallest lateral displacement makes it slide to one side.

Question from the audience. Can more than one geodesic connect the same two points, and can one of them be unstable? ^a

Response. Yes. The hill supplies exactly that example. Geodesic means stationary, not unique and not necessarily stable. The summit path satisfies the geodesic equation even though nearby paths can be shorter; unrestricted variations also show why stationarity should not be confused with a global maximum or minimum.

^aLecture timestamp: 00:22:09–00:25:14.

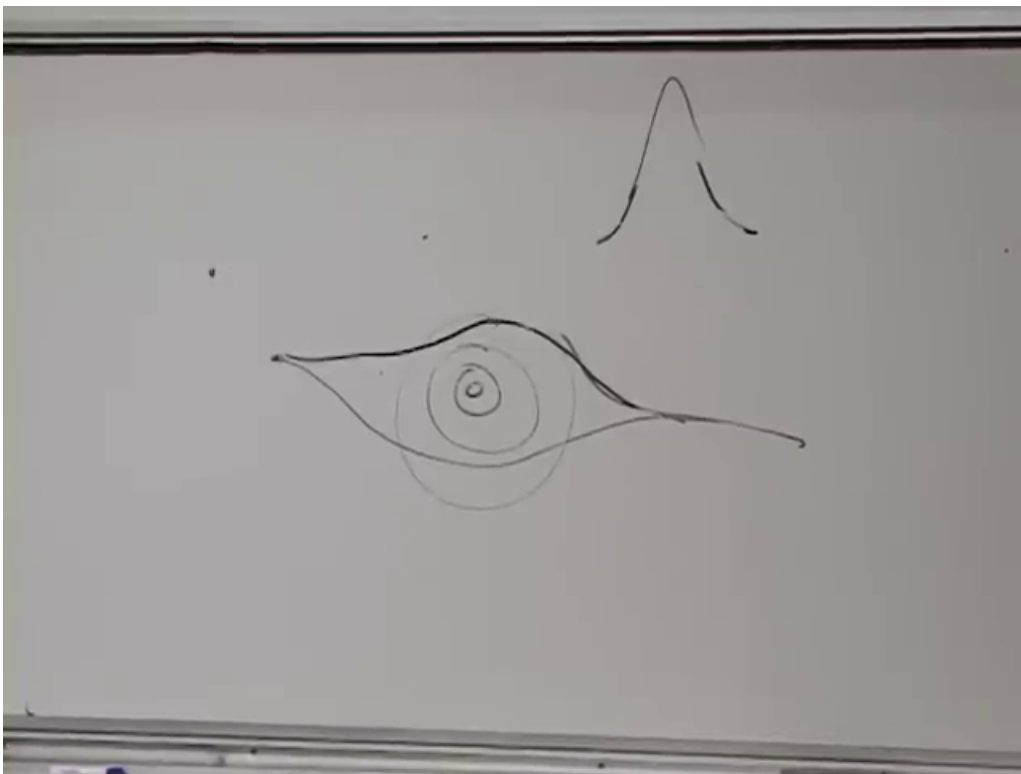


Figure 1.5: Three stationary paths around a hill: two stable routes around the sides and one unstable route over the summit.

Lecture frame: 00:23:55.

For a free massive particle the action is proper time,

$$S = -mc^2 \int d\tau = -mc \int ds. \quad (1.10)$$

Stationarity of this action gives Eq. (1.9). Thus free fall, stationary proper time, and the usual least-action machinery are three descriptions of the same motion.

Question from the audience. Is the proper-time principle different from the ordinary principle of least action? ^a

Response. No. Proper time supplies the Lagrangian for a free relativistic particle. Applying the Euler–Lagrange equations to that action yields the geodesic equation in any metric.

^aLecture timestamp: 00:25:14–00:27:10.

1.4.1 Using coordinate time

Take $x^0 = ct$ and define $v^i = dx^i/dt$. The line element is

$$c^2 d\tau^2 = g_{00} c^2 dt^2 + 2g_{0i} c dt dx^i + g_{ij} dx^i dx^j. \quad (1.11)$$

Therefore

$$S = \int L dt, \quad L = -mc^2 \sqrt{g_{00} + \frac{2g_{0i}v^i}{c} + \frac{g_{ij}v^i v^j}{c^2}}. \quad (1.12)$$

Question from the audience. Why does the mixed g_{0i} term carry a factor of two? ^a

Response. Because the symmetric quadratic form contains both $g_{0i} dx^0 dx^i$ and $g_{i0} dx^i dx^0$. Metric symmetry makes them equal, so together they give $2g_{0i} dx^0 dx^i$.

^aLecture timestamp: 00:29:09–00:29:23.

The Euler–Lagrange equations derived with t are equivalent to the geodesic equation, but t is generally not an affine parameter. Changing from τ to t uses

$$\frac{d}{d\tau} = \frac{dt}{d\tau} \frac{d}{dt}$$

and introduces a term proportional to dx^μ/dt :

$$\frac{d^2 x^\mu}{dt^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} = f(t) \frac{dx^\mu}{dt}. \quad (1.13)$$

The function f reflects the choice of a nonaffine parameter; it does not represent a new force.

Question from the audience. Why not compare the Euler–Lagrange equation and the geodesic equation term by term immediately? ^a

Response. They initially use different parameters. The standard geodesic equation uses affine proper time, while the displayed Lagrangian uses coordinate time. After the chain-rule conversion the two descriptions agree; carrying every algebraic term adds length but no new principle.

^aLecture timestamp: 00:32:26–00:36:37.

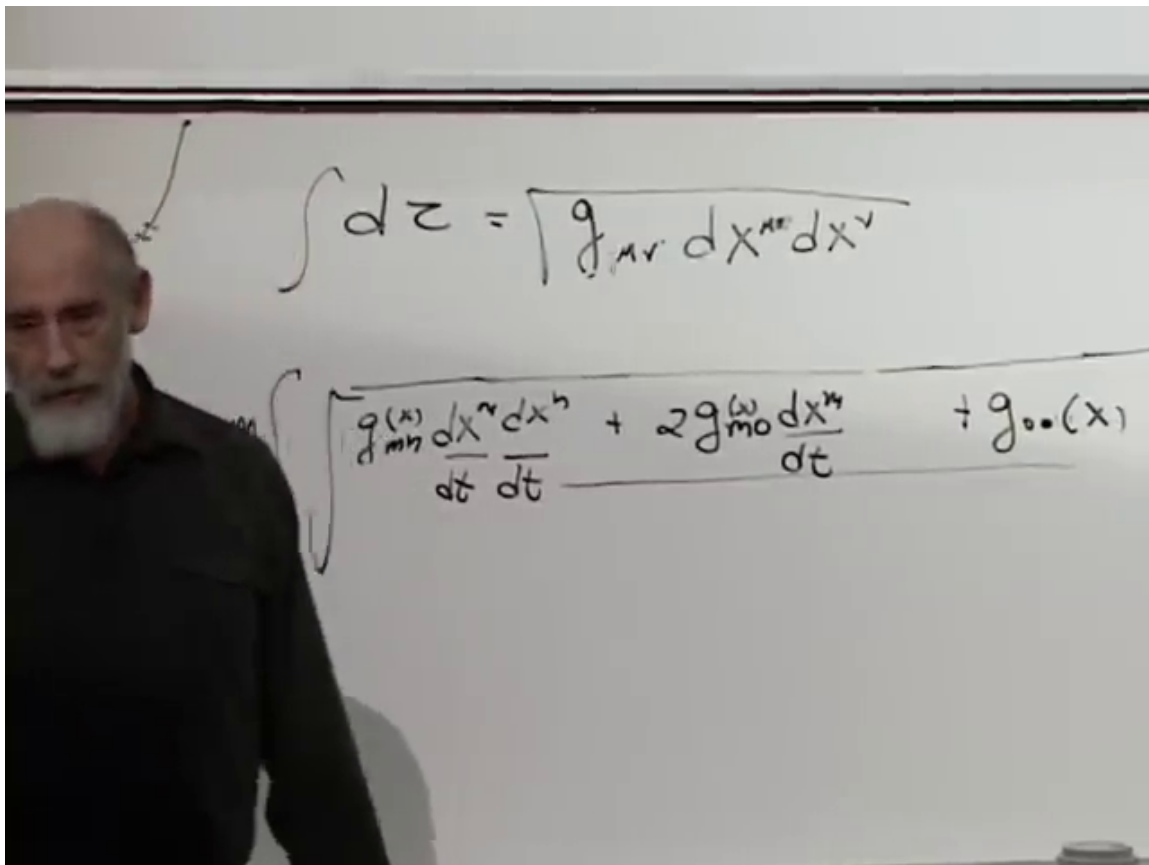


Figure 1.6: The proper-time action rewritten as a coordinate-time Lagrangian with the g_{00} , g_{0i} , and g_{ij} terms visible inside the square root.

Lecture frame: 00:31:48.

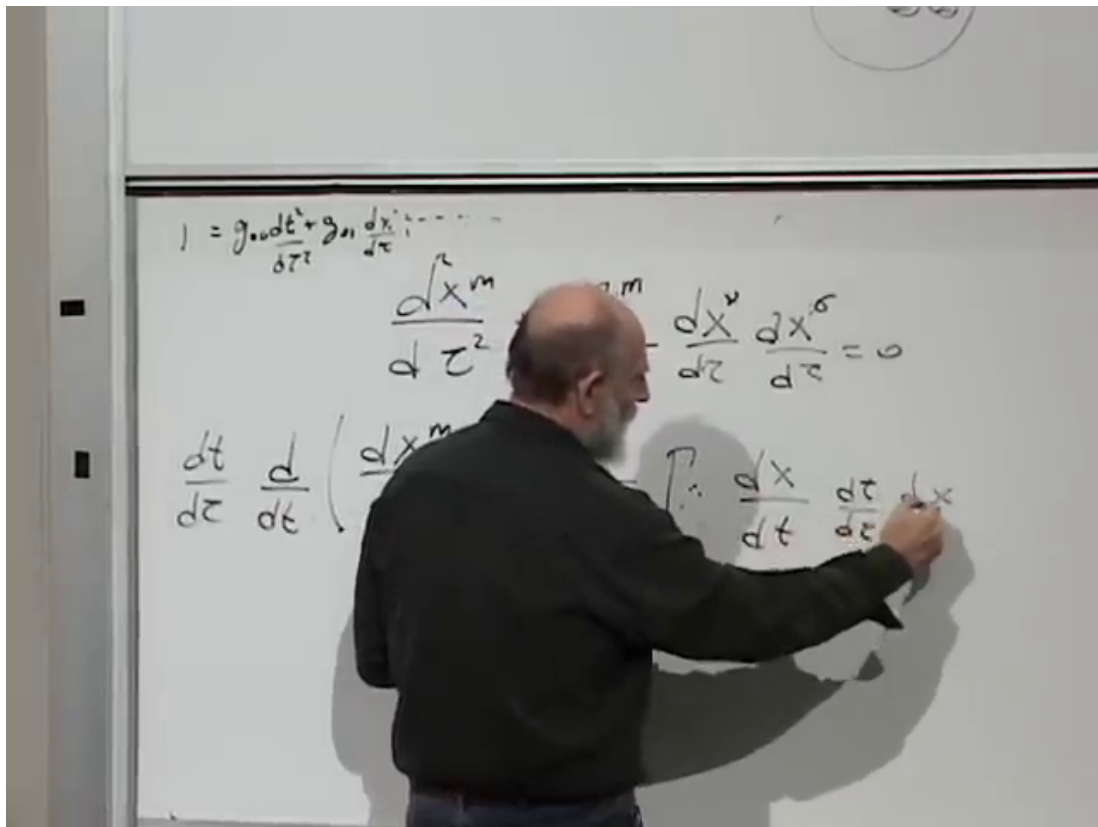


Figure 1.7: The board conversion between derivatives with respect to proper time and coordinate time in the geodesic equation.

Lecture frame: 00:36:02.

1.5 The Newtonian Limit Lives in g_{00}

Assume a static weak field, no appreciable mixed components, and slow motion:

$$\begin{aligned} g_{00} &= 1 + h_{00}, & g_{0i} &= 0, \\ g_{ij} &\simeq -\delta_{ij}, & |h_{00}| &\ll 1, \quad v \ll c. \end{aligned} \quad (1.14)$$

Then Eq. (1.12) becomes

$$L = -mc^2 \sqrt{1 + h_{00} - \frac{v^2}{c^2}}.$$

Using $\sqrt{1 + \epsilon} \simeq 1 + \epsilon/2$,

$$L \simeq -mc^2 + \frac{1}{2}mv^2 - \frac{1}{2}mc^2 h_{00}. \quad (1.15)$$

The first term is an irrelevant constant for the equations of motion. Comparison with

$$L_N = \frac{1}{2}mv^2 - m\Phi$$

gives

$$\boxed{h_{00} = \frac{2\Phi}{c^2}, \quad g_{00} \simeq 1 + \frac{2\Phi}{c^2}.} \quad (1.16)$$

Question from the audience. Why is the flat spatial contribution negative in the square root? ^a

Response. That is the declared Lorentzian signature. In flat spacetime $g_{00} = 1$ and $g_{ij} = -\delta_{ij}$, so $c^2 d\tau^2 = c^2 dt^2 - d\mathbf{x}^2$.

^aLecture timestamp: 00:37:04–00:37:44.

Question from the audience. Is the gravitational potential literally g_{00} ? ^a

Response. No. The potential is the small departure of g_{00} from its asymptotic constant, with $h_{00} = 2\Phi/c^2$. Adding a constant to Φ corresponds to a normalization of time and does not alter the Newtonian force.

^aLecture timestamp: 00:39:30–00:43:46.

1.6 Uniform Acceleration in Special Relativity

We now approach black holes indirectly. A Newtonian trajectory with constant coordinate acceleration,

$$X(T) = X_0 + V_0 T + \frac{1}{2} a T^2,$$

eventually has $|dX/dT| > c$, so it cannot describe uniform acceleration for all time in special relativity. Constant *proper* acceleration instead traces the hyperbola

$$X^2 - c^2 T^2 = \rho^2. \quad (1.17)$$

Its asymptotes are the null lines $X = \pm cT$. A Lorentz boost moves a point along the same hyperbola, just as an ordinary rotation moves a point around a circle.

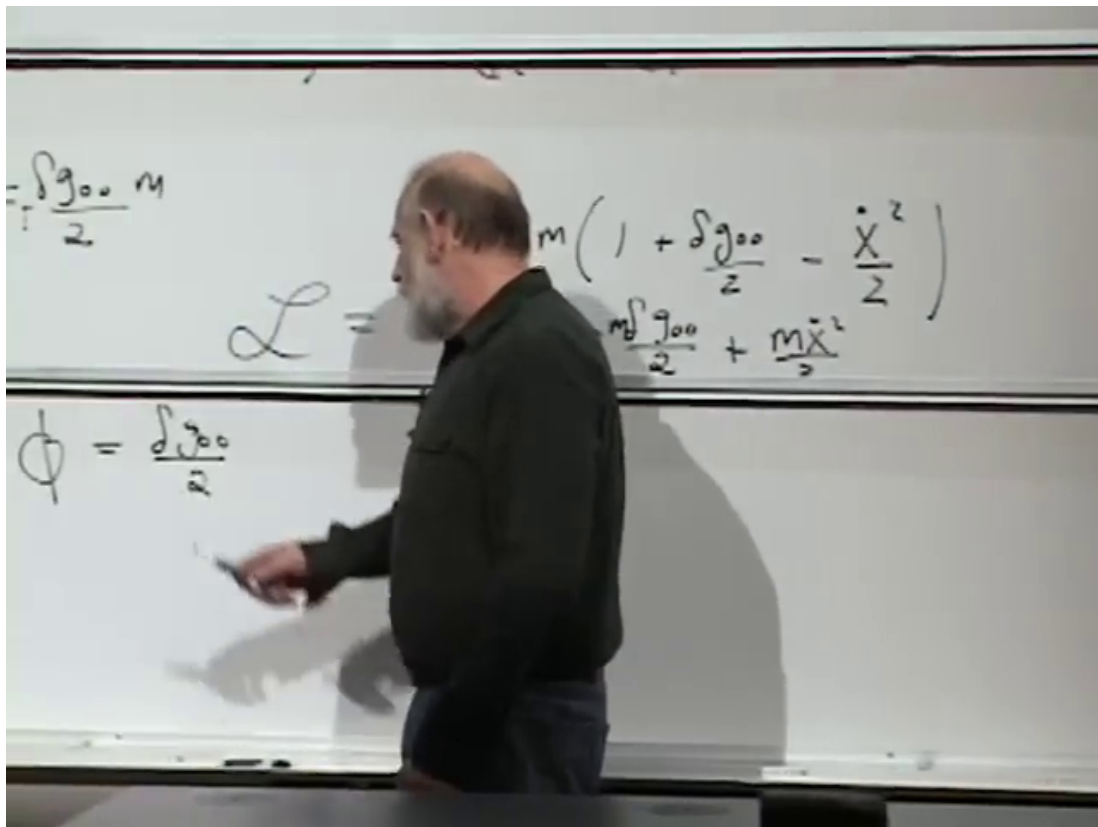


Figure 1.8: The weak-field square-root expansion and its comparison with kinetic minus potential energy.

Lecture frame: 00:43:35.

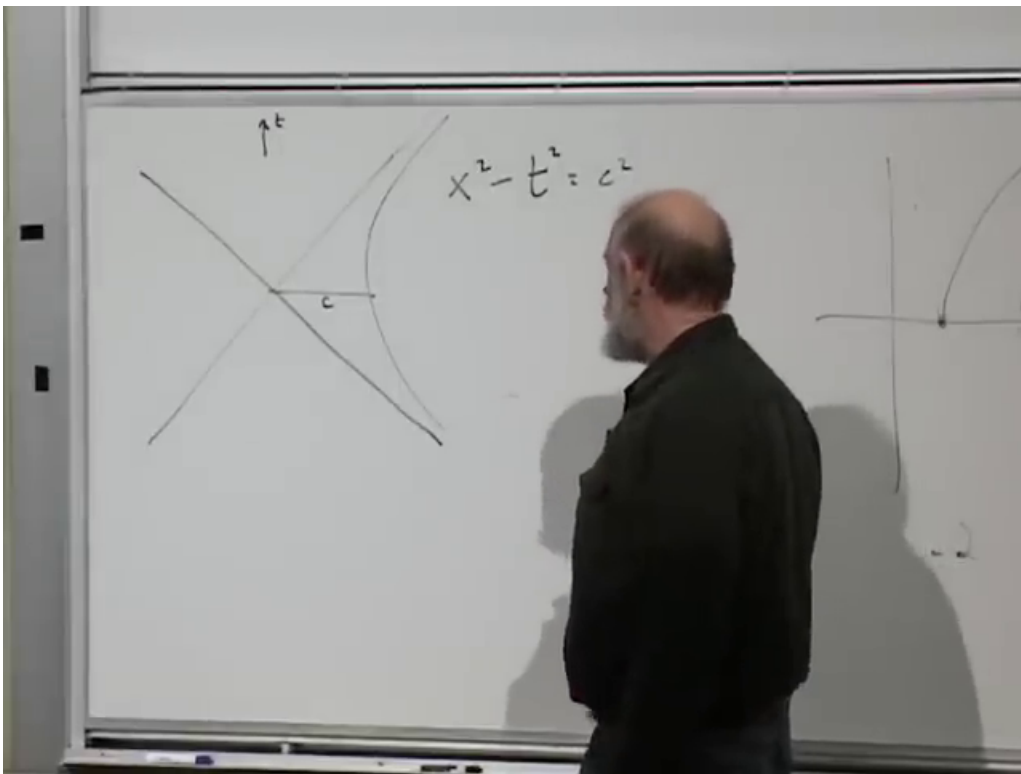


Figure 1.9: A uniformly accelerated worldline as a hyperbola inside its light-cone asymptotes, contrasted on the board with the Newtonian parabola.

Lecture frame: 00:49:40.

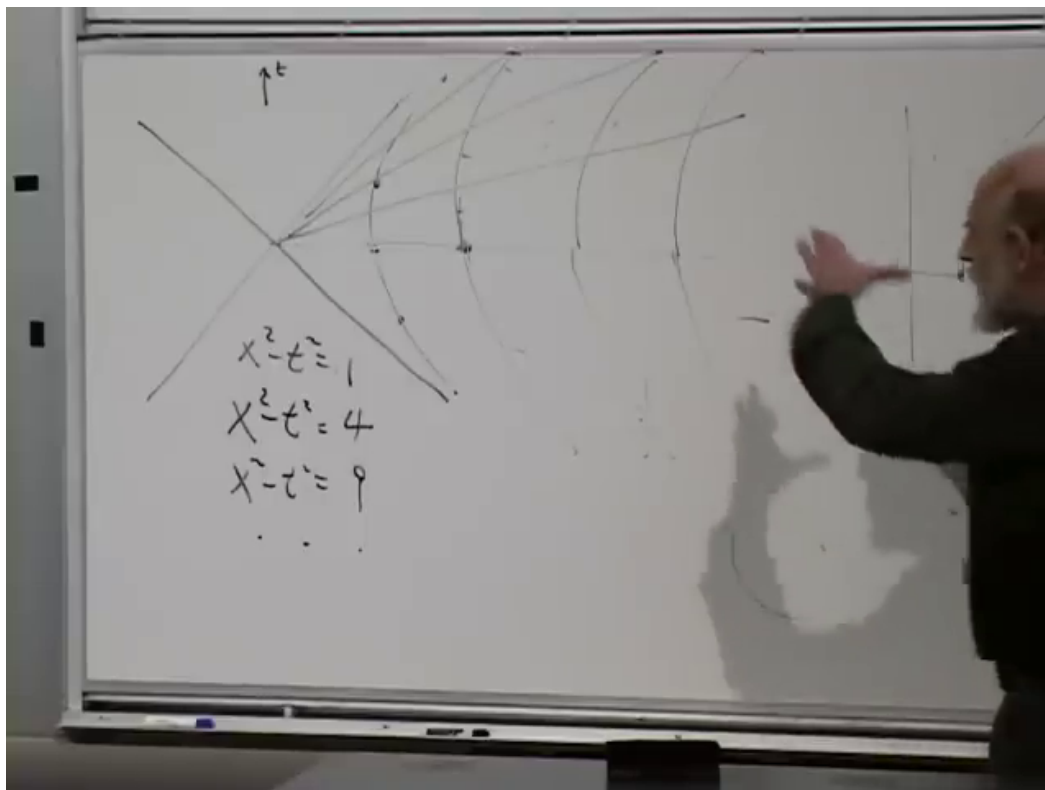


Figure 1.10: A Rindler congruence: constant- ρ hyperbolae and tilted surfaces of constant rapidity inside the right wedge.

Lecture frame: 00:54:58.

Question from the audience. Why is the relativistic uniformly accelerated path a hyperbola rather than the familiar parabola? ^a

Response. Uniformity refers to acceleration measured in the instantaneous rest frame. The invariant hyperbola preserves that proper acceleration under boosts, while its coordinate speed approaches c without exceeding it. The Newtonian parabola has no such speed limit.

^aLecture timestamp: 00:47:24–00:52:22.

Parameterize the hyperbola by a dimensionless rapidity η :

$$X = \rho \cosh \eta, \quad cT = \rho \sinh \eta. \quad (1.18)$$

Along $\rho = \text{constant}$,

$$d\tau = \frac{\rho}{c} d\eta, \quad a(\rho) = \frac{c^2}{\rho}.$$

A family of constant- ρ hyperbolae forms a congruence of accelerated observers. Lines of constant η are their successive surfaces of simultaneity.

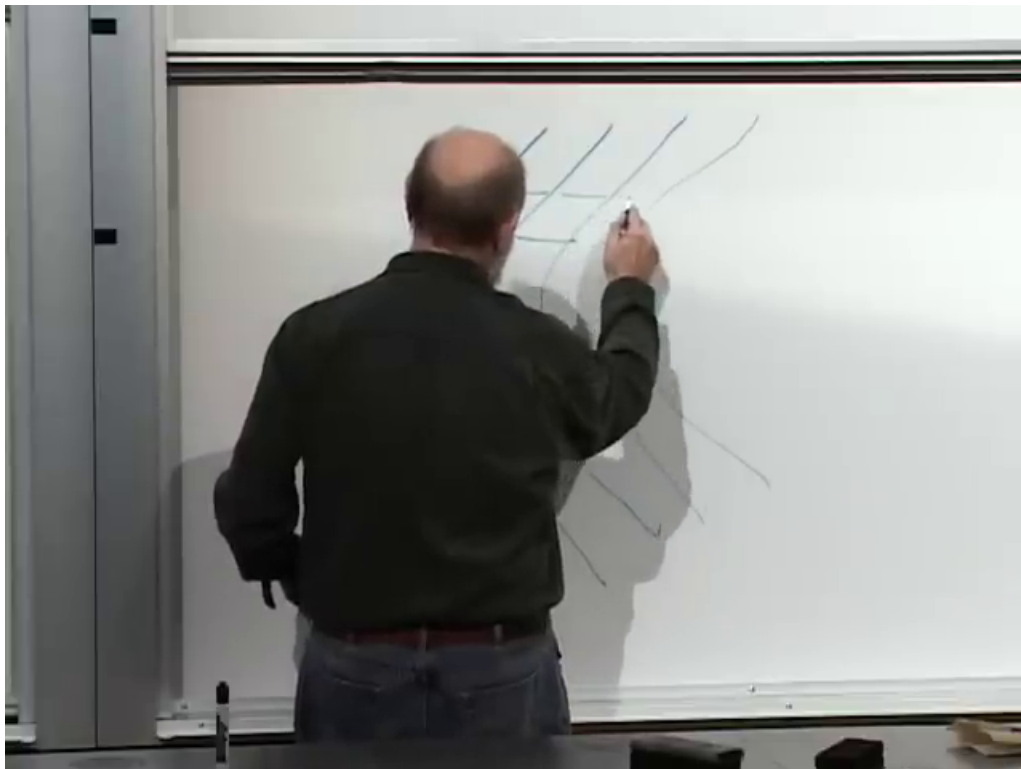


Figure 1.11: The acceleration gradient across the congruence. The hyperbolae crowd toward the null boundary, where the required proper acceleration diverges.

Lecture frame: 00:58:12.

Question from the audience. Do all observers in a uniformly accelerated frame feel the same proper acceleration? ^a

Response. No. A Born-rigid family requires $a(\rho) = c^2/\rho$. Observers nearer the null boundary accelerate more strongly. Simply translating one hyperbola to make equal accelerations would not preserve the proper separation of the observers.

^aLecture timestamp: 00:55:22–00:58:30.

The equivalence principle lets the accelerated observers describe their experience as a static gravitational field. Near a chosen worldline $\rho = \rho_0$, put $\rho = \rho_0 + z$ and define local time $ct_0 = \rho_0\eta$. The Rindler lapse then gives

$$g_{00} = \left(1 + \frac{z}{\rho_0}\right)^2 \simeq 1 + \frac{2z}{\rho_0}.$$

Equation (1.16) identifies

$$\Phi \simeq \frac{c^2 z}{\rho_0} = a_0 z, \quad a_0 = \frac{c^2}{\rho_0}.$$

The effective field strengthens toward smaller ρ . Nothing in the underlying spacetime is singular; the noninertial observers are using rockets in flat Minkowski space.

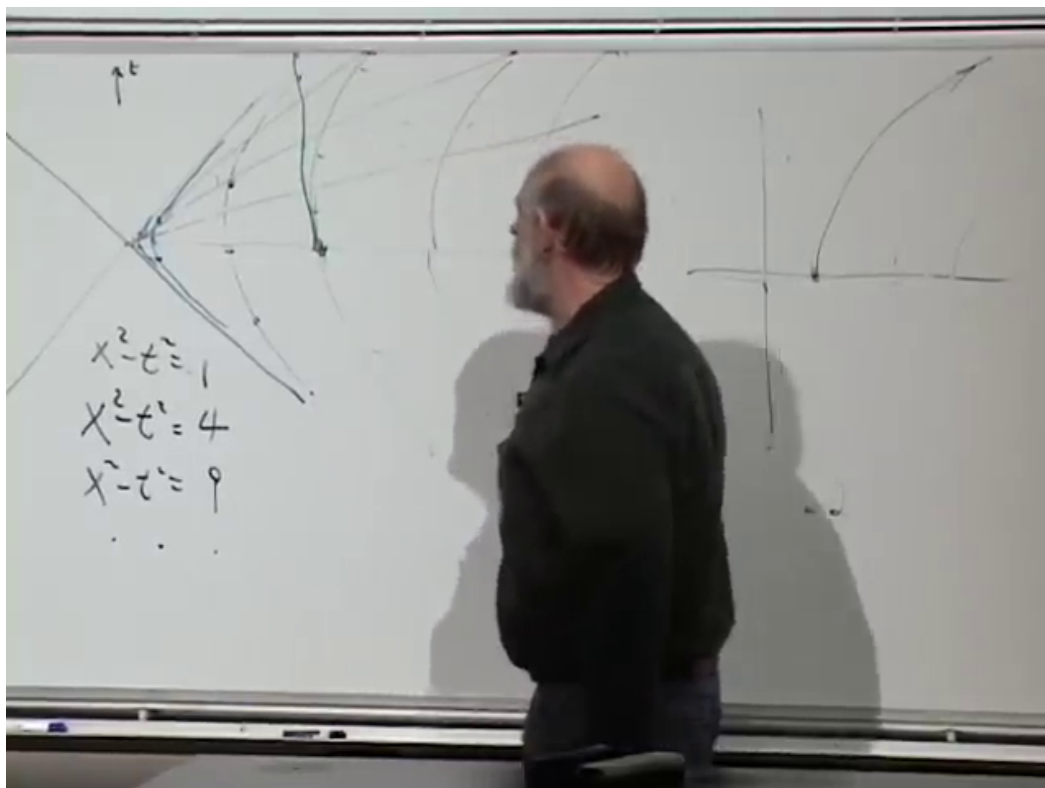


Figure 1.12: An inertial rock crosses the accelerated congruence on a straight worldline while the Rindler observers follow hyperbolae.

Lecture frame: 01:02:38.

1.7 A Horizon Without Curvature

Release a rock from one accelerated observer. The rock ceases to fire its rocket and follows an inertial straight line in the Minkowski diagram. The accelerated congruence bends away from that line, so its members describe the rock as falling toward $\rho = 0$.

Question from the audience. What happens to a rock released from an accelerated observer?
^a

Response. It follows an ordinary inertial geodesic. In Minkowski coordinates nothing pulls it; in the accelerated frame it falls toward the Rindler horizon. The two descriptions concern the same worldline.

^aLecture timestamp: 01:02:05–01:03:42.

Imagine that the rock flashes, rings, or broadcasts periodically. Signals emitted after it crosses the future null line $X = cT$ cannot reach any observer who remains forever in the right Rindler wedge. That null line is the observers' future horizon. It is a causal boundary, not a material membrane and not a region of large curvature.

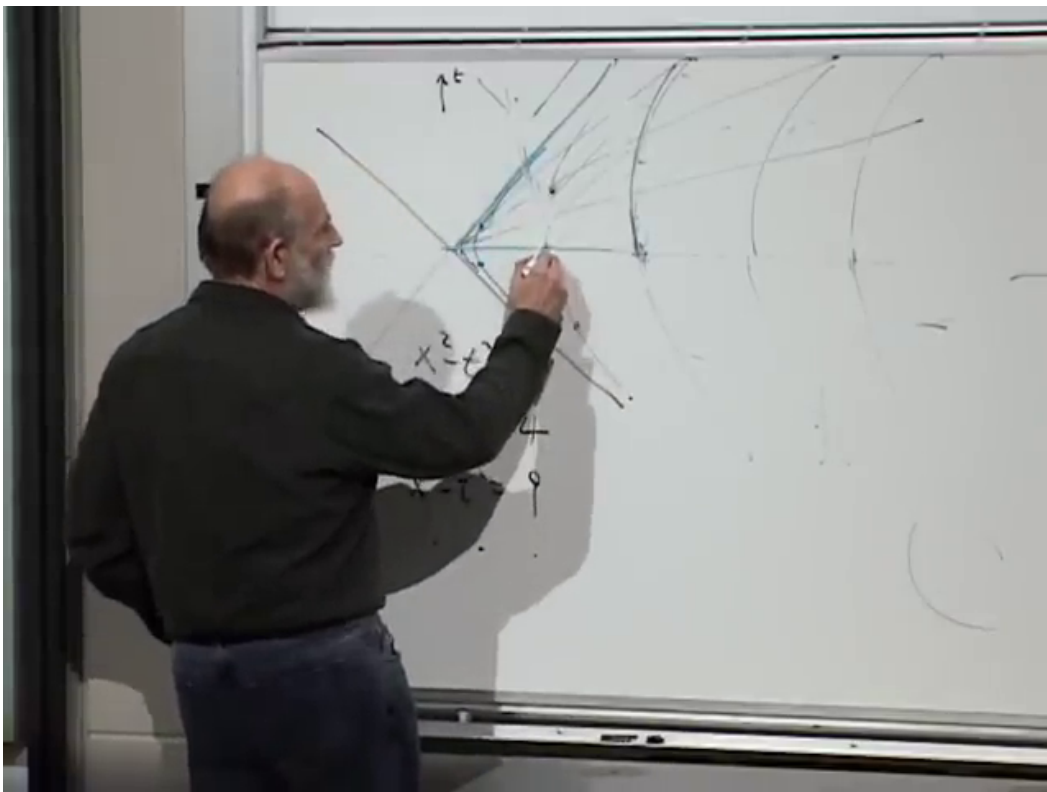


Figure 1.13: Light signals from a released rock and the null boundary beyond which they can no longer catch a persistently accelerated observer.

Lecture frame: 01:06:18.

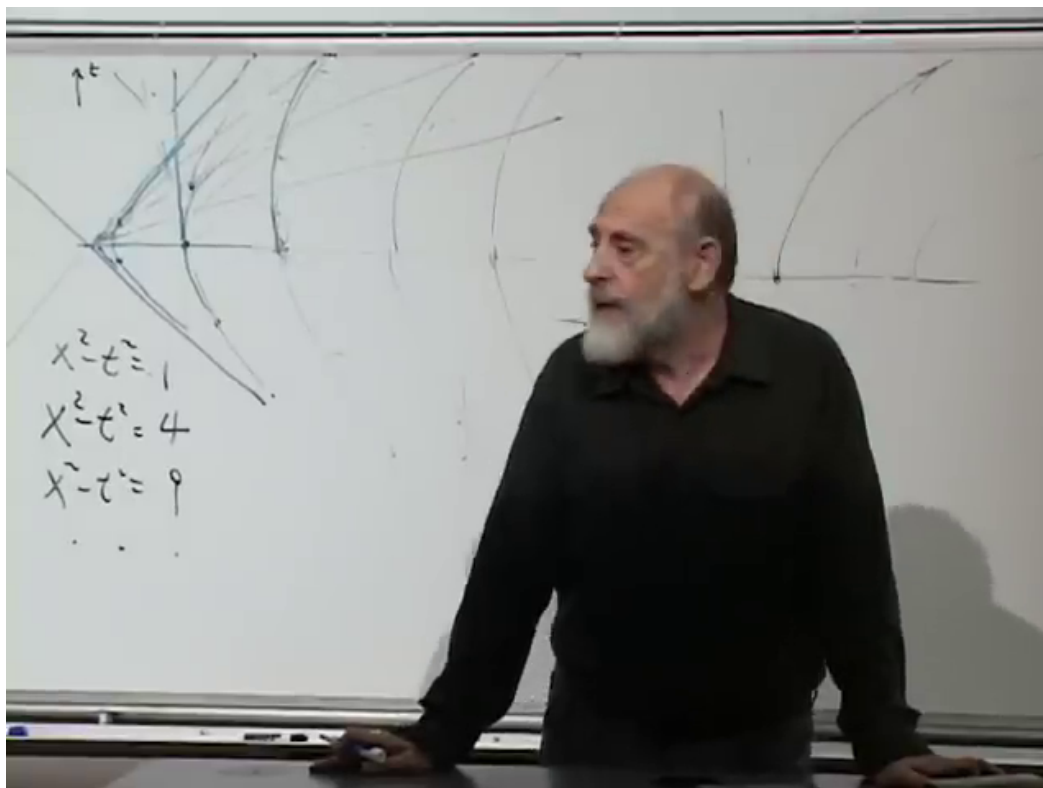


Figure 1.14: A later release produces a different inertial tangent, but the same causal lesson: the rock crosses the Rindler horizon while accelerated observers remain in the wedge.

Lecture frame: 01:08:02.

Question from the audience. Is the horizon an absolute place, and is the accelerated observer forbidden to cross it? ^a

Response. No. It is tied to the observer's decision to accelerate forever. The observer may switch off the rocket, follow an inertial path, and cross the same null line. Different accelerated worldlines have correspondingly shifted causal boundaries.

^aLecture timestamp: 01:06:53–01:07:49.

If a second rock is released later, its inertial path is tangent to the hyperbola at the later release event. It too crosses the null boundary, even though its straight line differs from the first.

Question from the audience. Does releasing the rock later prevent it from crossing the horizon? ^a

Response. No. The release changes which inertial tangent the rock follows. Every such released trajectory can pass through the future null boundary while the eternally accelerated observers remain outside it.

^aLecture timestamp: 01:07:50–01:08:19.

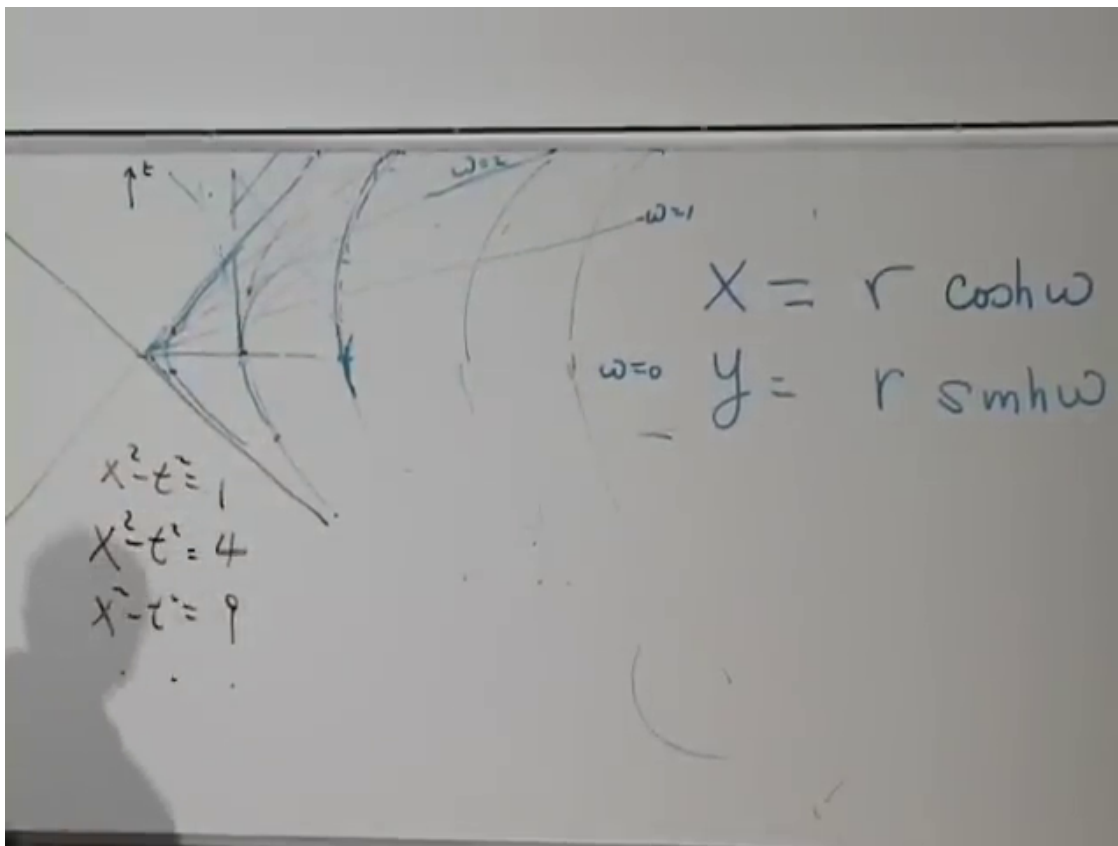


Figure 1.15: Circular polar coordinates beside their hyperbolic counterpart $X = \rho \cosh \eta$, $cT = \rho \sinh \eta$.

Lecture frame: 01:13:40.

Question from the audience. After the rock has crossed, can it still receive signals from the accelerated observers? ^a

Response. Yes. Future-directed light from outside can cross inward and reach the rock. The forbidden direction is outward: signals emitted by the rock after crossing cannot return to the eternally accelerated observers. The horizon is a one-way causal boundary, not a reflecting wall.

^aLecture timestamp: 01:08:19–01:09:05.

1.8 Rindler Coordinates

Equation (1.18) is the Lorentzian analogue of polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

The circular angle becomes a hyperbolic angle. The radial coordinate $\rho > 0$ labels a hyperbola, and $-\infty < \eta < \infty$ labels the observer's position along it. The transverse coordinates y, z remain unchanged.

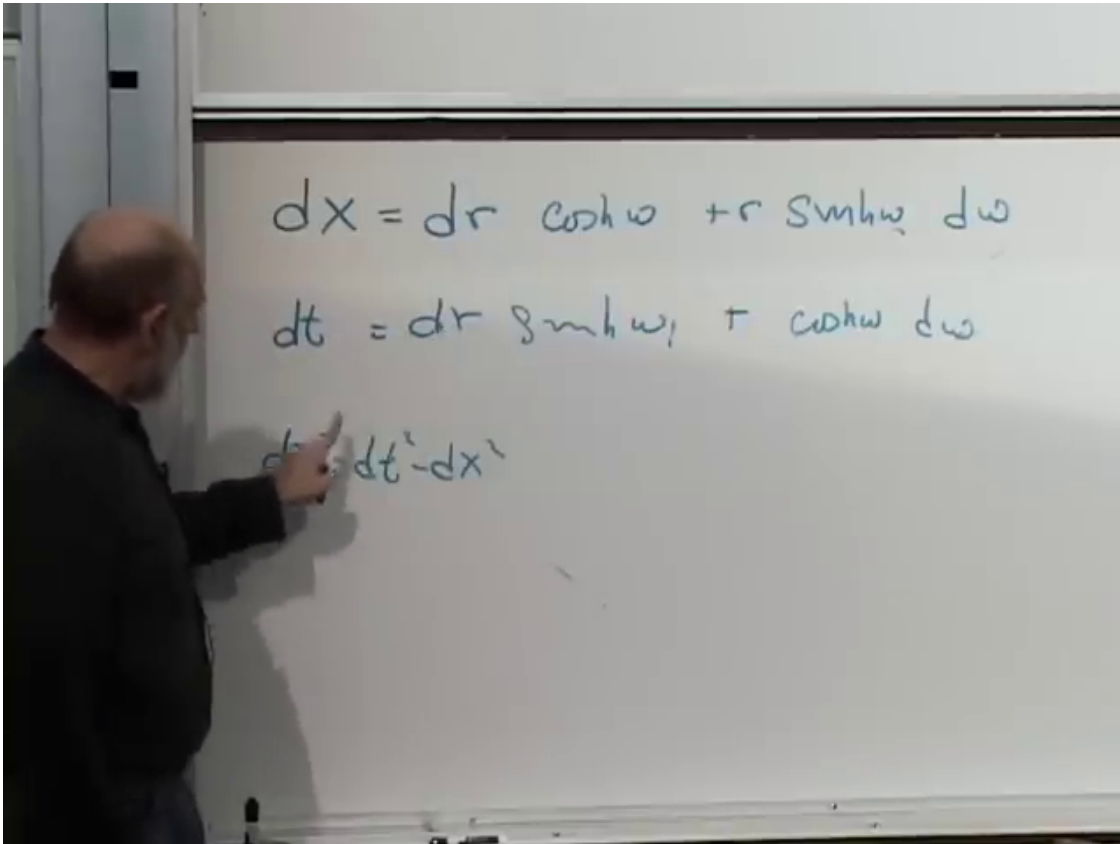


Figure 1.16: The differentiated hyperbolic coordinate transformation before substitution into the Minkowski line element.

Lecture frame: 01:18:50.

Differentiate the transformation:

$$dX = \cosh \eta d\rho + \rho \sinh \eta d\eta, \quad (1.19)$$

$$c dT = \sinh \eta d\rho + \rho \cosh \eta d\eta. \quad (1.20)$$

Substitution into

$$ds^2 = c^2 dT^2 - dX^2 - dy^2 - dz^2$$

cancels the cross terms. Since $\cosh^2 \eta - \sinh^2 \eta = 1$, the result is

$$\boxed{ds^2 = \rho^2 d\eta^2 - d\rho^2 - dy^2 - dz^2.} \quad (1.21)$$

Question from the audience. Does Rindler space cover all of Minkowski spacetime? ^a

Response. No. With $\rho > 0$, these coordinates cover only the right wedge $X > |cT|$. The coordinate boundary $\rho = 0$ corresponds to the two null rays that form the past and future Rindler horizons.

^aLecture timestamp: 01:24:03–01:24:20.

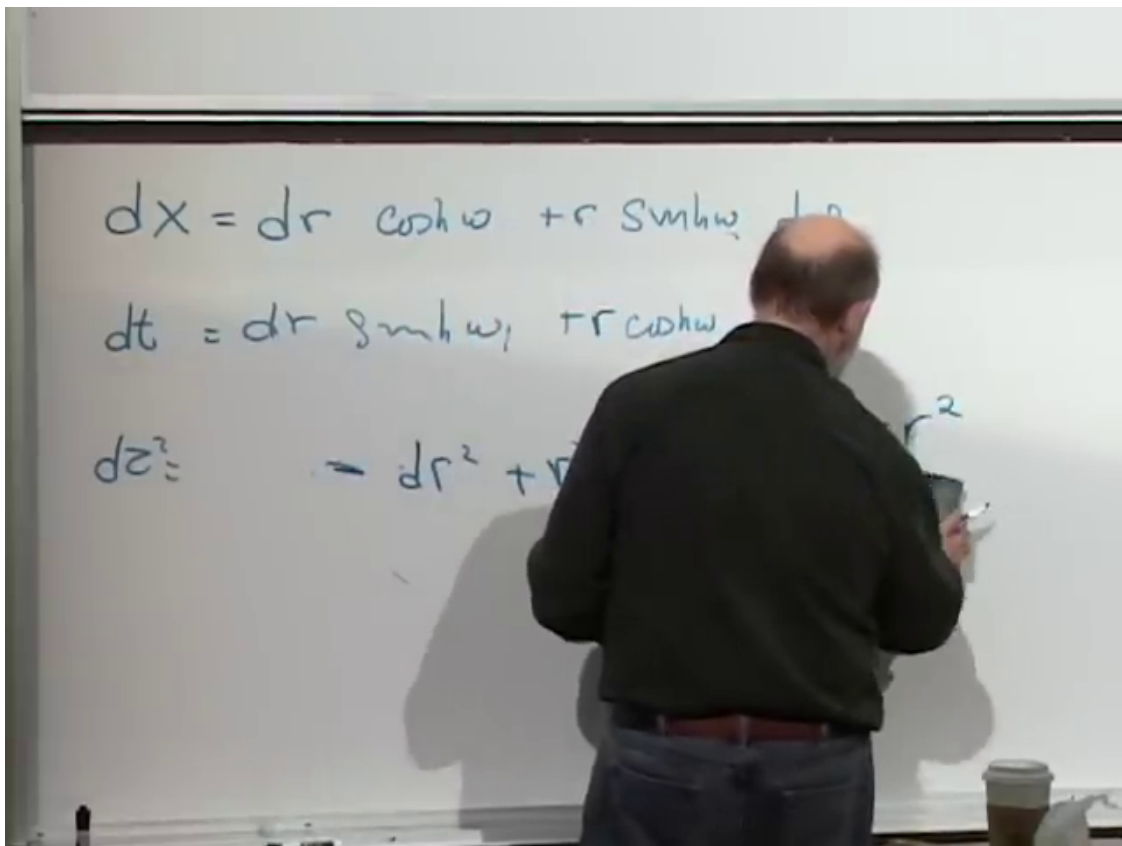


Figure 1.17: The completed Rindler metric. The coordinate components are $g_{\eta\eta} = \rho^2$ and $g_{\rho\rho} = -1$, even though the underlying spacetime is flat.

Lecture frame: 01:22:05.

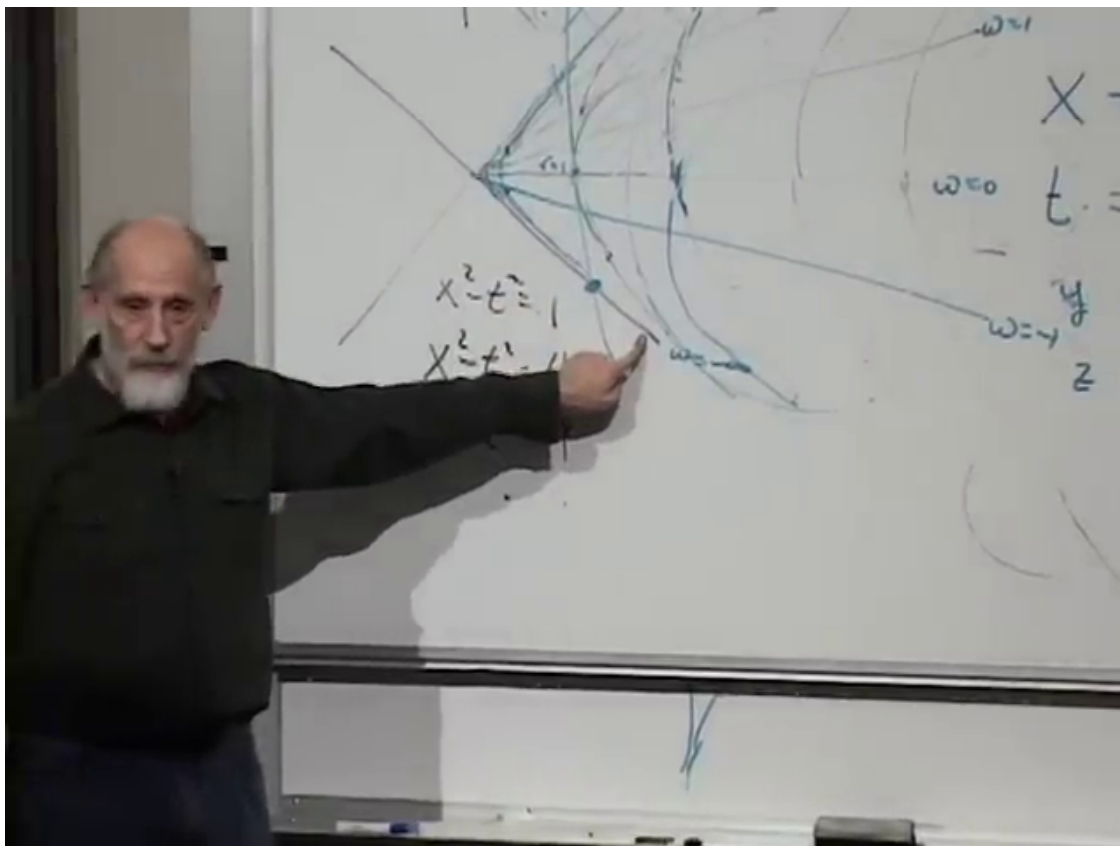


Figure 1.18: The same inertial straight line viewed against the Rindler hyperbolae. In accelerated coordinates it appears to come from the past horizon, turn around at maximum ρ , and approach the future horizon.

Lecture frame: 01:27:05.

1.8.1 An inertial geodesic in accelerated coordinates

Take the inertial worldline $X = X_0 > 0$. Using Eq. (1.18),

$$\rho(\eta) = \frac{X_0}{\cosh \eta}. \quad (1.22)$$

As $\eta \rightarrow -\infty$, the line enters the wedge through $\rho = 0$; at $\eta = 0$ it reaches its maximum $\rho = X_0$; and as $\eta \rightarrow +\infty$ it returns to $\rho = 0$. The horizon crossings occur at finite inertial proper time but at infinite Rindler time.

Question from the audience. How does a freely falling path look in the (ρ, η) coordinates? ^a

Response. It appears at $\rho = 0$ when $\eta = -\infty$, rises to a maximum distance from the horizon, and returns to $\rho = 0$ when $\eta = +\infty$. The infinite coordinate time is a property of the accelerated clock, not an infinite proper time for the falling object.

^aLecture timestamp: 01:24:22–01:28:32.

Successive pulses from the falling rock take longer to catch the accelerated observer and arrive increasingly redshifted. The observer therefore sees the rock slow and fade near the horizon. The rock itself crosses the null line without encountering curvature or a local singularity. It can still

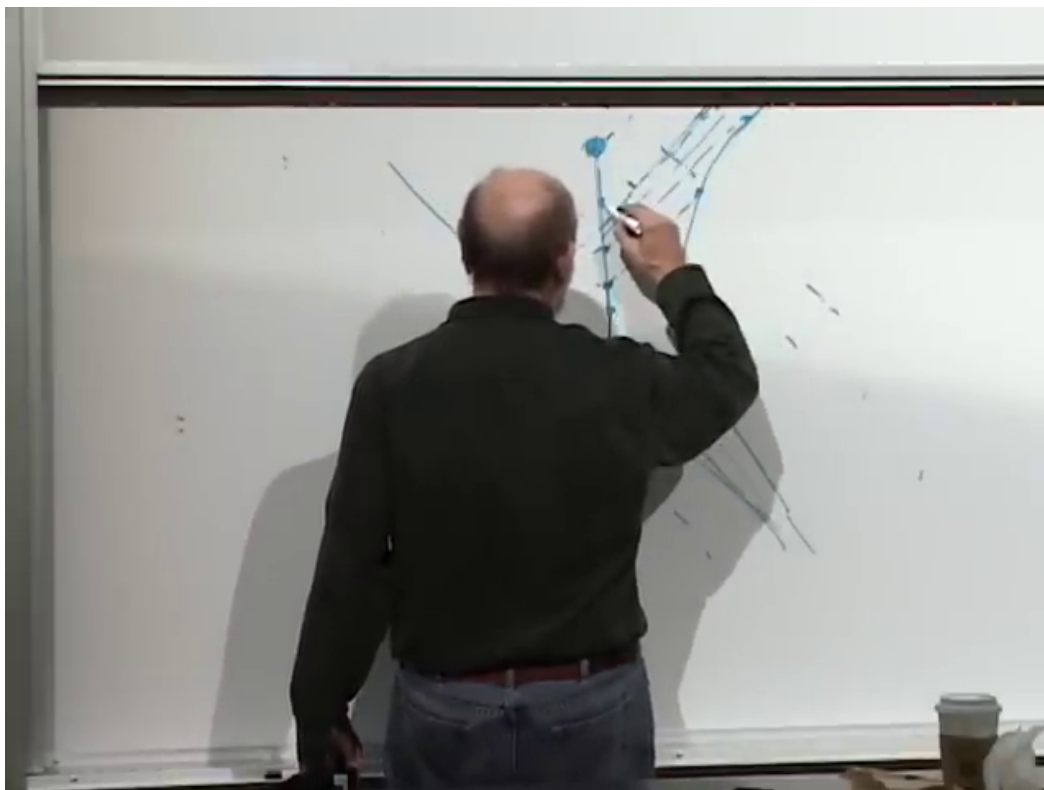


Figure 1.19: The completed causal sketch near the Rindler horizon. The accelerated observer's outgoing view and the infaller's continued reception of inward signals are not symmetric.

Lecture frame: 01:33:55.

receive inward signals from the exterior; their measured frequencies depend on the relative boost and on the direction of propagation, so one should not replace the causal statement by a universal redshift or blueshift slogan.

Question from the audience. What does the accelerated observer see, and what does the falling rock experience at the horizon? ^a

Response. The accelerated observer receives an ever slower, redder sequence of signals and assigns the crossing to $\eta = +\infty$. The rock crosses in finite proper time and notices no local singularity. After crossing it may receive signals sent inward from the exterior, but it cannot send a causal signal back to the eternally accelerated observer.

^aLecture timestamp: 01:28:33–01:34:20.

1.9 Spherical Symmetry and the Exterior Problem

The Rindler construction required no curvature. We now turn to a genuinely gravitating, spherically symmetric source. The lecture does not solve the full field equations component by component. It invokes the result appropriate to the exterior:

Theorem 1.1 (Birkhoff). *Every spherically symmetric solution of the vacuum Einstein equations with vanishing cosmological constant is locally a member of the Schwarzschild family. In particular, the vacuum exterior of a bounded spherical source is static and depends only on its total mass.*

Question from the audience. Why not write every component of Einstein's equation and solve the whole system on the board? ^a

Response. Spherical symmetry and the vacuum equations have already reduced the answer to the Schwarzschild exterior. A component-by-component derivation is lengthy and would obscure the geometric issue the lecture now wants to study: the horizon. The time-time coefficient can be motivated from Newtonian gravity; the radial coefficient is then stated as the field-equation result.

^aLecture timestamp: 01:34:23–01:37:31.

Use the standard angular convention: θ is polar angle from the north pole and ϕ is azimuth. The unit two-sphere has

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (1.23)$$

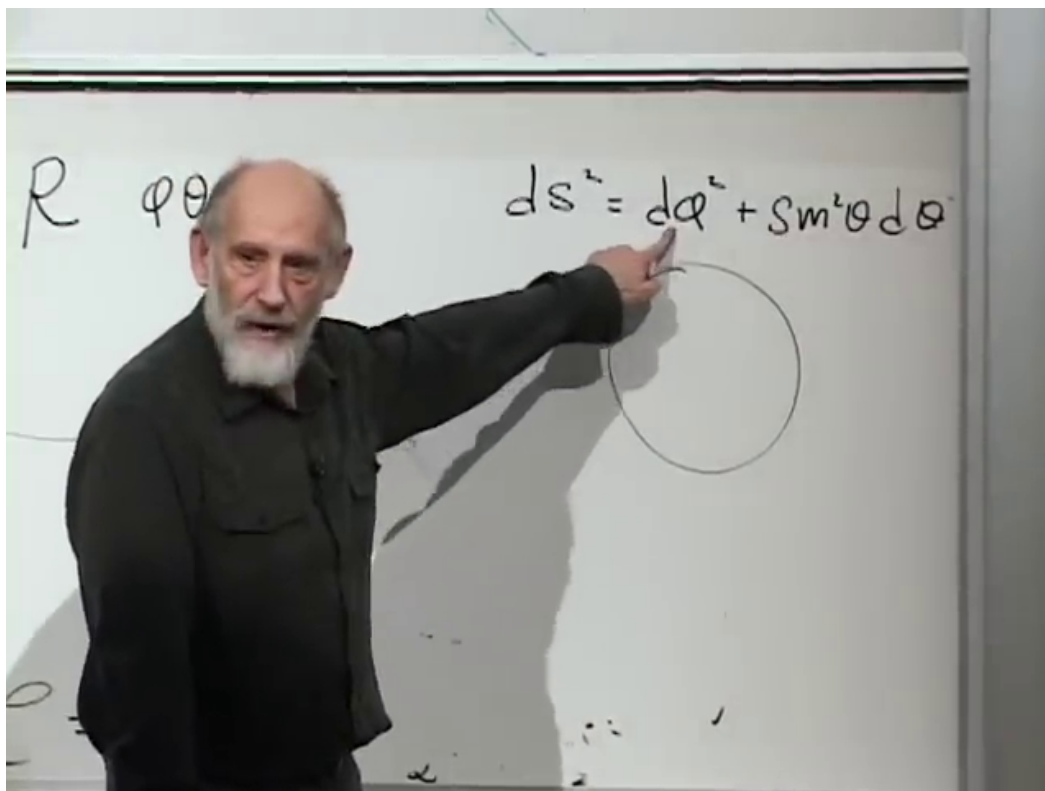


Figure 1.20: The unit-sphere angular element after the live notation is normalized to the standard polar angle θ and azimuth ϕ .

Lecture frame: 01:41:55.

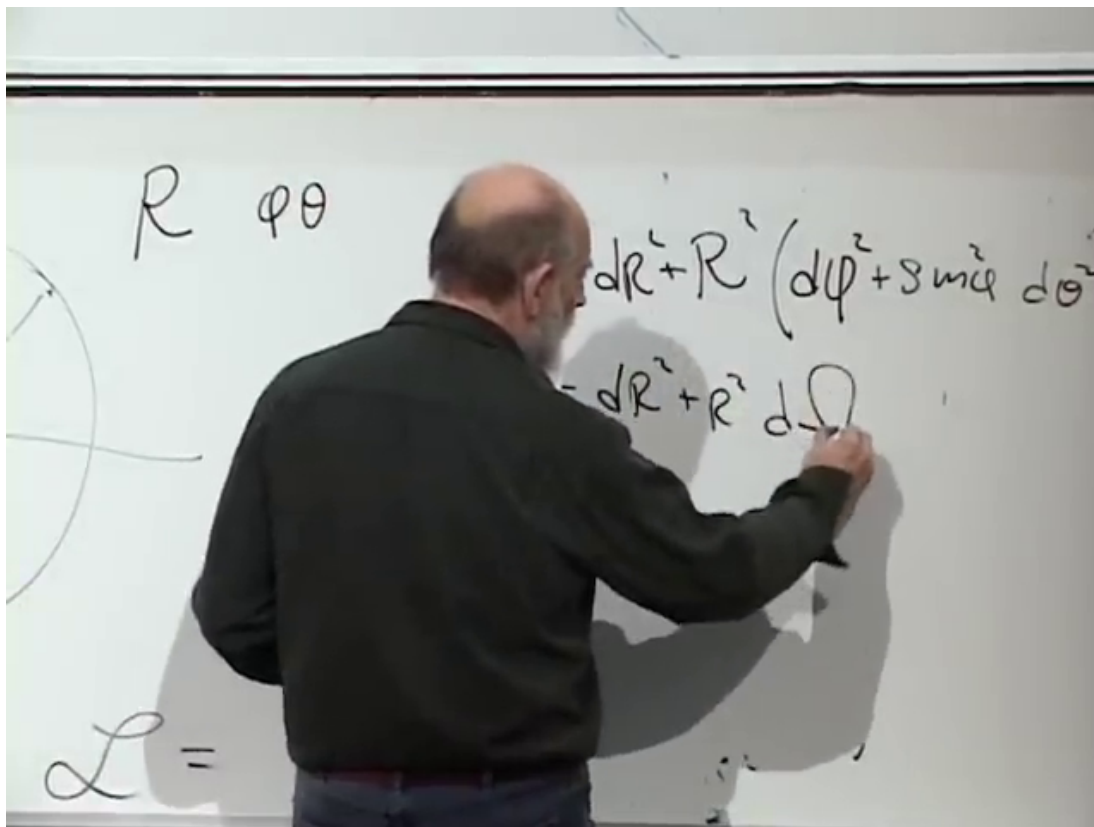


Figure 1.21: The radial and angular pieces of the flat three-dimensional line element.

Lecture frame: 01:44:05.

Question from the audience. Which angular coordinate carries the sine factor? ^a

Response. With the conventional choice used here, θ measures polar angle and ϕ measures azimuth, so $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. Interchanging the names is harmless if done consistently; the geometry, not the letter, is what matters.

^aLecture timestamp: 01:40:32–01:42:24.

Flat three-dimensional space in spherical coordinates is

$$d\ell^2 = dr^2 + r^2 d\Omega^2. \quad (1.24)$$

The coordinate r is the areal radius: every orbit of the rotation group has area $4\pi r^2$.

Adding time gives the flat spacetime metric

$$ds^2 = c^2 dt^2 - dr^2 - r^2 d\Omega^2. \quad (1.25)$$

1.10 The Schwarzschild Metric

Outside a spherical mass M , the Newtonian potential is

$$\Phi(r) = -\frac{GM}{r}. \quad (1.26)$$

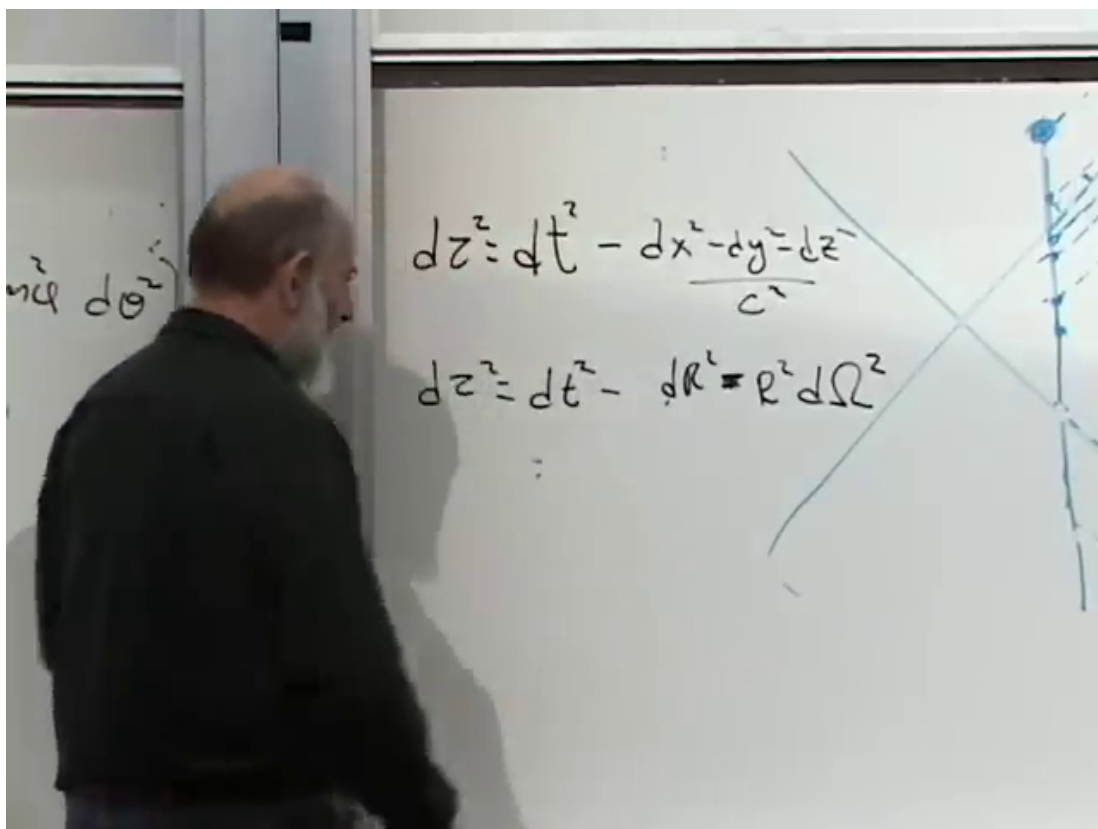


Figure 1.22: Minkowski spacetime written in coordinates adapted to spherical symmetry.

Lecture frame: 01:46:15.

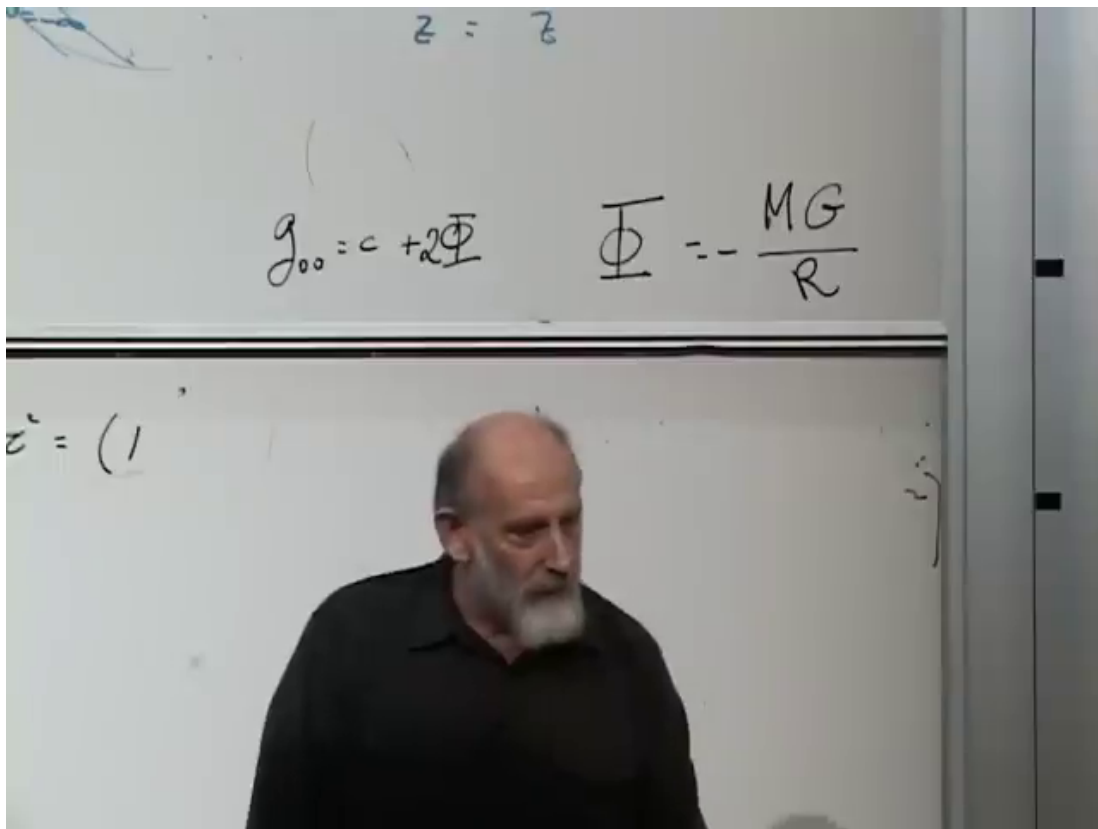


Figure 1.23: The Newtonian potential and the weak-field relation that fixes the leading Schwarzschild time component.

Lecture frame: 01:49:42.

The weak-field relation Eq. (1.16) then requires

$$g_{00} \simeq 1 - \frac{2GM}{c^2 r}. \quad (1.27)$$

The constant 1 enforces asymptotic flatness. Far from the source, the metric approaches Eq. (1.25).

Define the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}. \quad (1.28)$$

The exact vacuum exterior metric is

$$ds^2 = \left(1 - \frac{r_s}{r}\right) c^2 dt^2 - \frac{dr^2}{1 - r_s/r} - r^2 d\Omega^2. \quad (1.29)$$

The angular coefficient retains the form $-r^2 d\Omega^2$ because r has been chosen as areal radius. This is a coordinate choice, not a claim that angular geometry is unaffected by gravity. The reciprocal factor in the radial term cannot be obtained from the Newtonian potential alone; it is the additional information supplied by the vacuum Einstein equations.

Dimensional analysis makes clear why ordinary gravitational fields are weak. The expansion parameter is

$$\frac{GM}{c^2 r} = \frac{r_s}{2r}. \quad (1.30)$$

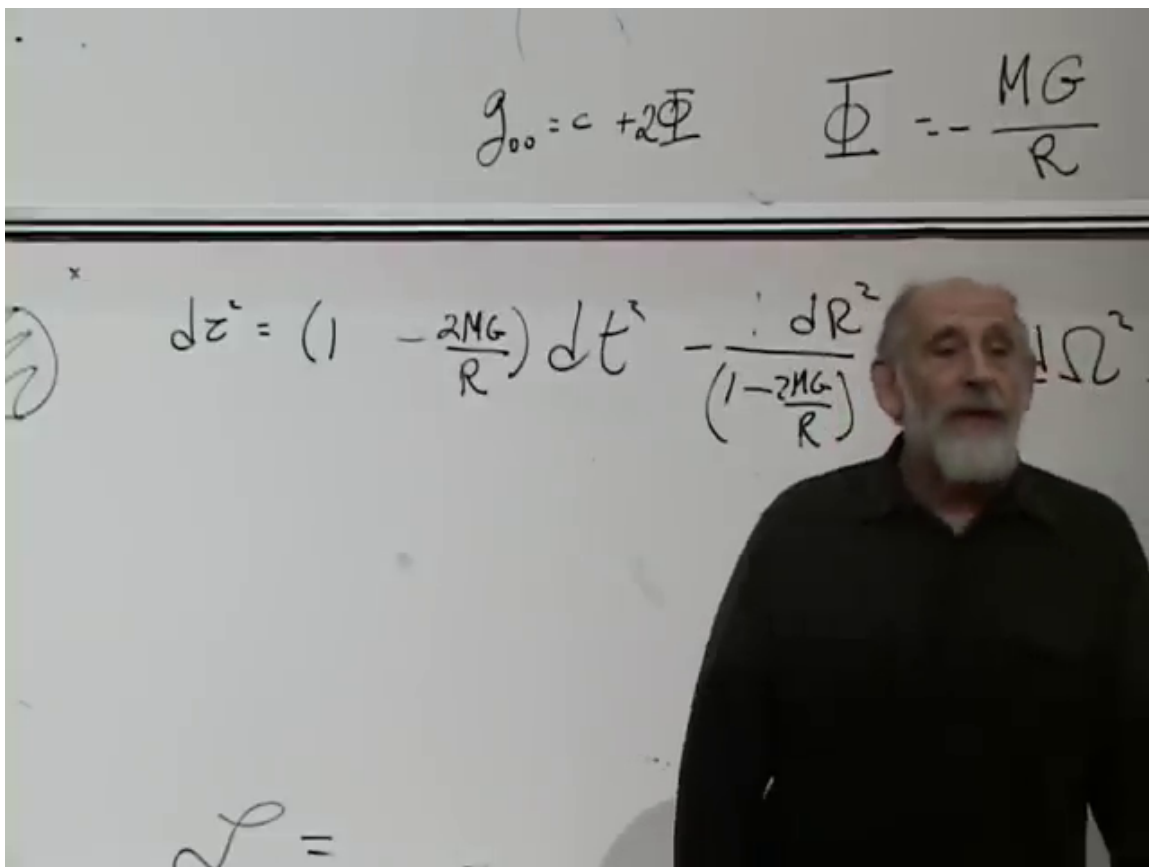


Figure 1.24: The Schwarzschild line element assembled on the board: the Newtonian time factor, its reciprocal radial factor, and the unchanged areal angular term.

Lecture frame: 01:53:20.

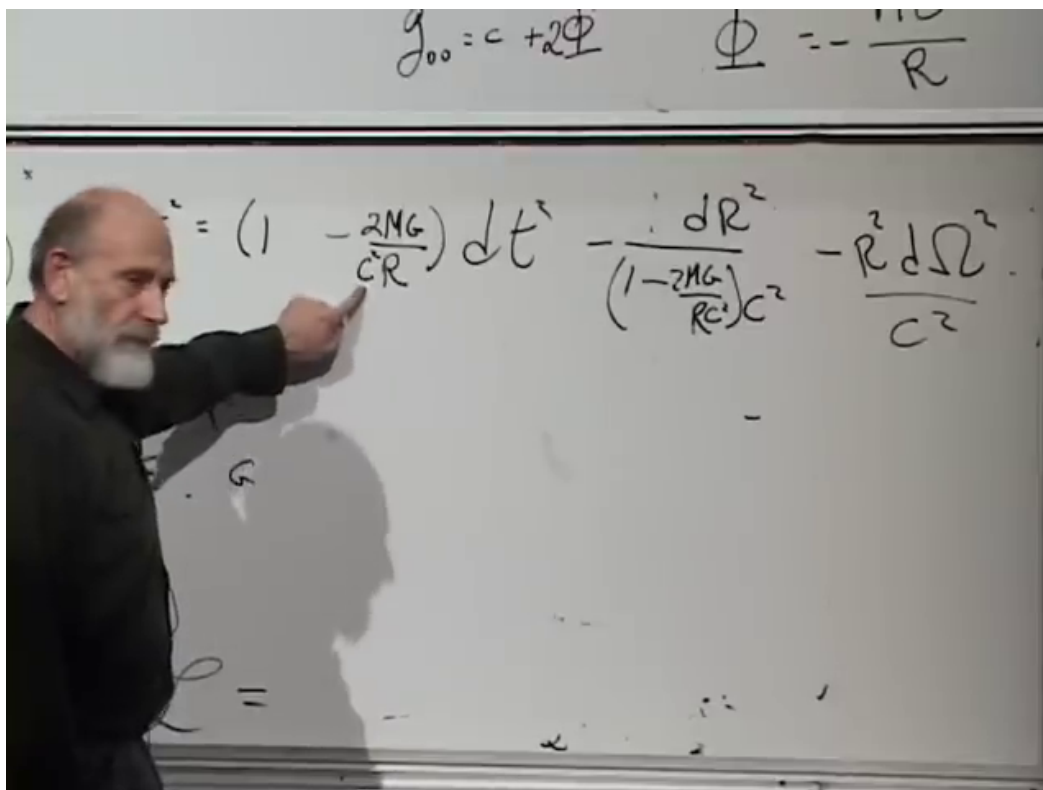


Figure 1.25: The completed Schwarzschild metric with the powers of c restored, making the dimensionless compactness $2GM/(c^2 r)$ explicit.

Lecture frame: 01:56:35.

For an ordinary star at radii much larger than r_s , this number is small. For a nonrotating, uncharged black hole, the surface $r = r_s$ is the horizon.

At $r = r_s$, the coefficient of dt^2 vanishes and the coefficient of dr^2 diverges in Schwarzschild coordinates. The lecture does not yet interpret this as a physical singularity. Its final promise is instead that, after a suitable near-horizon coordinate reorganization, the (t, r) part of the geometry will resemble Rindler space. The resemblance is hidden by the present coordinates and will be made explicit next.

1.11 From Rindler Space to a Black-Hole Horizon

Topology fixes integrated curvature, not the local metric. A torus admits a flat metric, a punctured sphere admits one pulled back from the plane, and no discontinuous digit encoding can replace a smooth coordinate transformation. Normal coordinates flatten g and its first derivatives at one event while leaving second derivatives.

The proper-time action gives free motion, and its weak-field limit identifies $g_{00} \simeq 1 + 2\Phi/c^2$. A family of uniformly accelerated observers occupies one wedge of flat spacetime and has a one-way horizon even though the Riemann tensor vanishes. Schwarzschild geometry then supplies a curved exterior. Its horizon looks different algebraically, but the causal and near-horizon comparison is prepared.